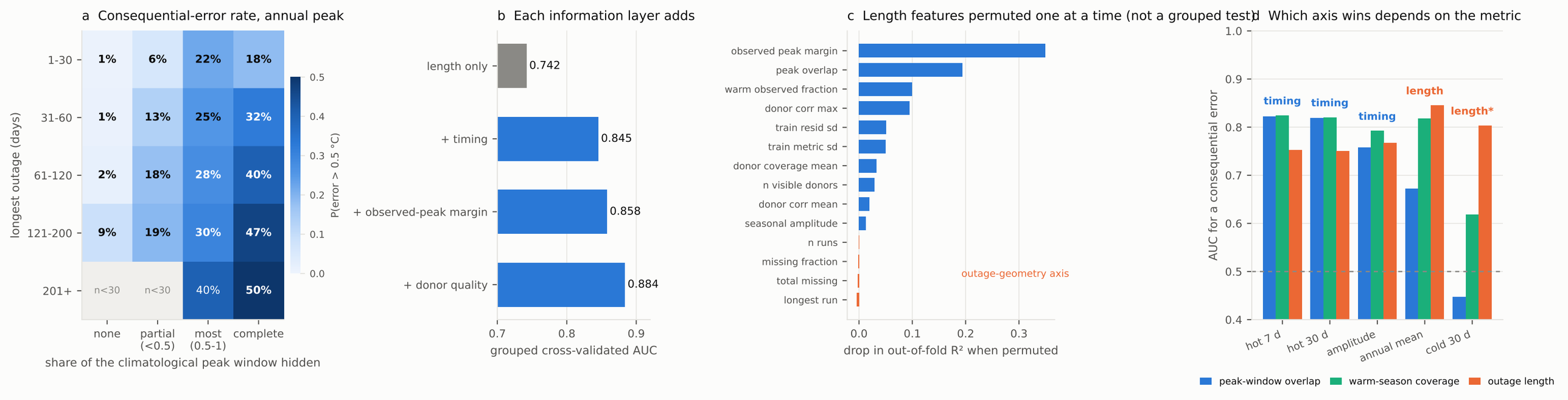
a Consequential-error rate, annual peak
longest outage (days)
1-30
31-60
61-120
121-200
201+
1%
6%
22%
18%
1%
13%
25%
32%
2%
18%
28%
40%
9%
19%
30%
47%
n<30
n<30
40%
50%
none
partial (<0.5)
most (0.5-1)
complete
share of the climatological peak window hidden
P(error > 0.5 °C)
0.5
0.4
0.3
0.2
0.1
0.0
b Each information layer adds
length only
0.742
+ timing
0.845
+ observed-peak margin
0.858
+ donor quality
0.884
0.7
0.8
0.9
grouped cross-validated AUC
c Length features permuted one at a time (not a grouped test)
observed peak margin
peak overlap
warm observed fraction
donor corr max
train resid sd
train metric sd
donor coverage mean
n visible donors
donor corr mean
seasonal amplitude
n runs
missing fraction
total missing
longest run
outage-geometry axis
0.0
0.1
0.2
0.3
drop in out-of-fold R² when permuted
d Which axis wins depends on the metric
AUC for a consequential error
1.0
0.9
0.8
0.7
0.6
0.5
0.4
timing
timing
timing
length
length*
hot 7 d
hot 30 d
amplitude
annual mean
cold 30 d
peak-window overlap
warm-season coverage
outage length